

15.10 (a) We can follow the same procedure as in Problem 14.22, and notice that the motion of the particle is not periodic. This can be found in Landau and Lifshitz, Classical Theory of Fields, section 70. So, I will not repeat the calculation here.

(b) From part (a), the radiation cross section is

$$\frac{dX(\omega)}{d\omega} = 2\pi \int_0^{+\infty} \frac{dI(\omega, b)}{d\omega} b db = \frac{16}{3} \frac{(ze a \omega)^2}{c^3} e^{-\pi \omega / \omega_0} \int_0^{+\infty} \left\{ \dots \right\} b db$$

Since $\varepsilon = \sqrt{1 + (b/a)^2}$, then $b = a\sqrt{\varepsilon^2 - 1}$, $db = a^2 \varepsilon d\varepsilon / \sqrt{\varepsilon^2 - 1}$, $b db = a \varepsilon d\varepsilon$. Now, the integral becomes

$$a^2 \int_1^{+\infty} \left\{ \left[K'_{i\omega/\omega_0} \left(\frac{\omega \varepsilon}{\omega_0} \right) \right]^2 + \frac{\varepsilon^2 - 1}{\varepsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega \varepsilon}{\omega_0} \right) \right]^2 \right\} \varepsilon d\varepsilon$$

Make another variable substitution with $\varepsilon = \frac{\omega_0}{\omega} x$, the integral becomes

$$a^2 \frac{\omega_0^2}{\omega^2} \int_{\omega/\omega_0}^{+\infty} \left\{ \left[K'_{i\omega/\omega_0}(x) \right]^2 + \left(1 - \frac{\omega^2/\omega_0^2}{x^2} \right) \left[K_{i\omega/\omega_0}(x) \right]^2 \right\} x dx$$

The first term can be evaluated with integration by parts.

$$\begin{aligned} \int_{\omega/\omega_0}^{+\infty} \left[K'_{i\omega/\omega_0}(x) \right]^2 x dx &= \int_{\omega/\omega_0}^{+\infty} K'_{i\omega/\omega_0}(x) x d(K_{i\omega/\omega_0}(x)) \\ &= x K_{i\omega/\omega_0}(x) K'_{i\omega/\omega_0}(x) \Big|_{\omega/\omega_0}^{+\infty} - \int_{\omega/\omega_0}^{+\infty} K_{i\omega/\omega_0}(x) d \left[x K'_{i\omega/\omega_0}(x) \right] \end{aligned}$$

Since the modified Bessel function of second kind satisfies the following ODE,

$$\frac{d}{dx} \left[x K'_\nu(x) \right] = \left(1 + \frac{\nu^2}{x^2} \right) K_\nu(x),$$

then we are left with the boundary terms only, which leads to

$$\begin{aligned} \frac{dX(\omega)}{d\omega} &= \frac{16}{3} \frac{(ze a \omega)^2}{c^3} e^{-\pi \omega / \omega_0} a^2 \frac{\omega_0^2}{\omega^2} \frac{\omega}{\omega_0} K_{i\omega/\omega_0} \left(\frac{\omega}{\omega_0} \right) \left[-K'_{i\omega/\omega_0} \left(\frac{\omega}{\omega_0} \right) \right] \\ &= \frac{16}{3} \frac{(ze a v)^2}{c^3} e^{-\pi \omega / \omega_0} \frac{\omega}{\omega_0} K_{i\omega/\omega_0} \left(\frac{\omega}{\omega_0} \right) \left[-K'_{i\omega/\omega_0} \left(\frac{\omega}{\omega_0} \right) \right] \end{aligned}$$

(c) For $w/w_0 \ll 1$, using the asymptotic form of the Bessel function, $K_\nu(x) \rightarrow -\log(\frac{x}{2})$, $x \rightarrow 0$,

and $K'_\nu(x) = -\frac{1}{x}$, we have

$$\frac{dX(w)}{dw} = \frac{16}{3} \frac{(zeav)^2}{c^3} \frac{w}{w_0} \left[-\log\left(\frac{w}{2w_0}\right) \right] \left[\frac{w_0}{w} \right] = \frac{16}{3} \frac{(zeav)^2}{c^3} \log\left(\frac{2w_0}{w}\right).$$

Since $a = Ze^2/mv$, then

$$\begin{aligned} \frac{dX(w)}{dw} &= \frac{16}{3} \frac{(Ze^2/mv)^2}{c^3} \log\left(\frac{2mv^3}{Ze^2w}\right) \\ &= \frac{16}{3} \frac{Z^2e^2}{c^3} \left(\frac{Z^2e^2}{m}\right)^2 \frac{1}{(\beta c)^2} \log\left(\frac{2mv^3}{Ze^2w}\right) = \frac{16}{3} \frac{Z^2e^2}{c} \left(\frac{Z^2e^2}{mc^2}\right)^2 \frac{1}{\beta^2} \log\left(\frac{2mv^3}{Ze^2w}\right), \end{aligned}$$

which agrees with Eq. (15.75), up to some constant factor.

For $w/w_0 \gg 1$, using the result that $K_\nu(x) \rightarrow \sqrt{\frac{\pi}{2x}} e^{-x}$, $x \rightarrow +\infty$, the limiting form is proportional to $\exp\{-(\pi + \nu)w/w_0\}$.

(d) See Prob. 14.22